An Ontology-Guided Approach for Enhanced Documentation of Cave Microbial Culture Collections

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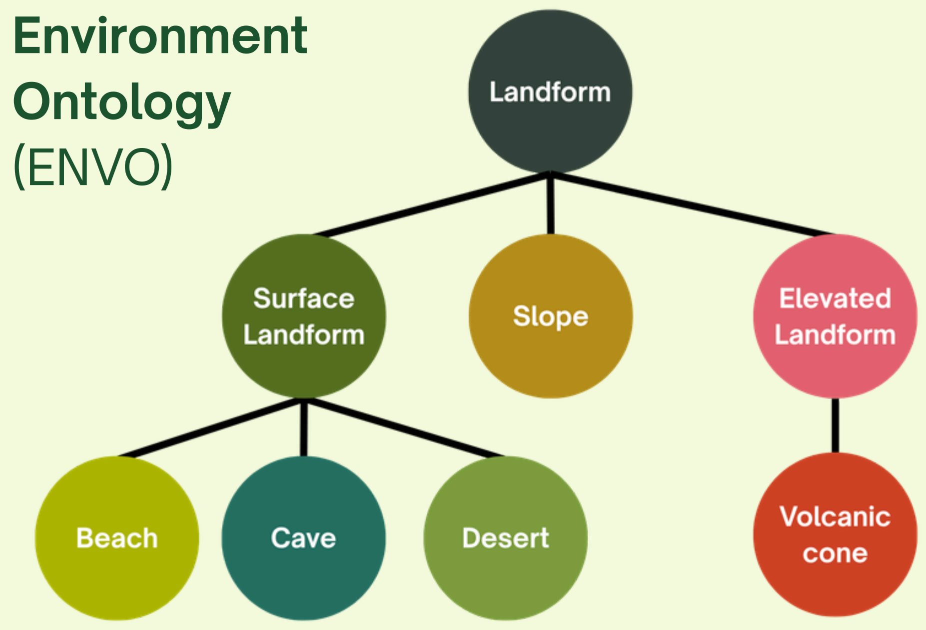
1 Introduction

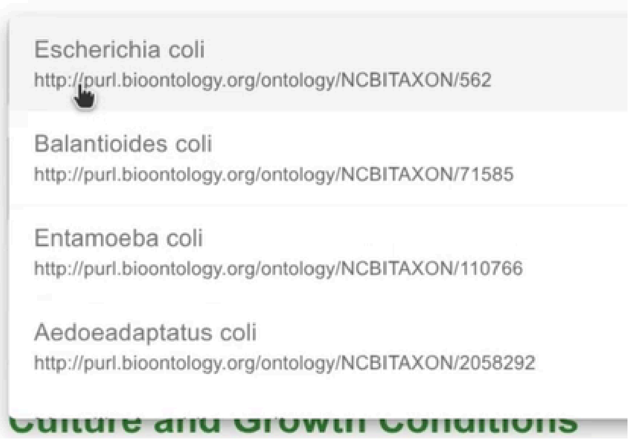
Free-text annotations in cave culture collection information systems lead to **inconsistent** descriptions of **identical** concepts, making cave microbial data difficult to interpret and compare.

low temperature === cold ?

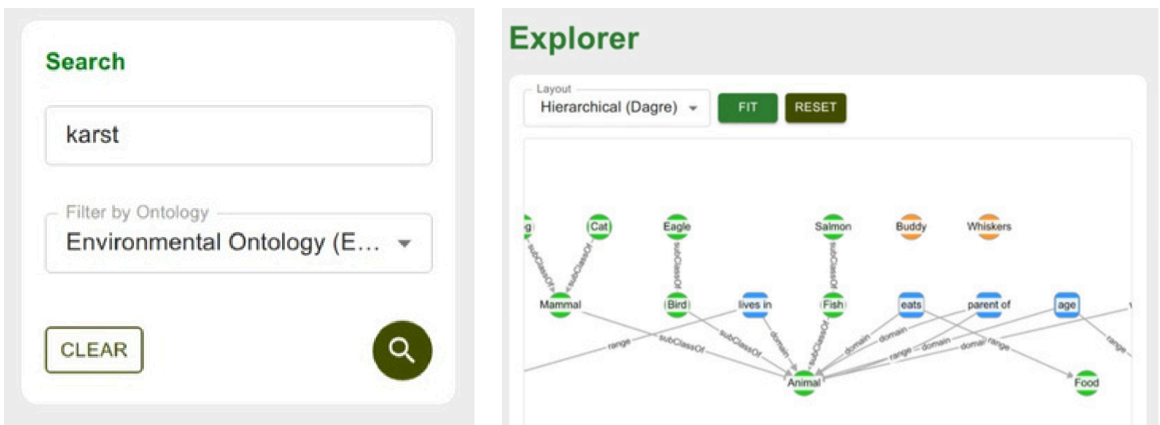
Ontologies bring a unique solution by providing **standardized terms** with defined relationships, promoting accuracy in data entry.



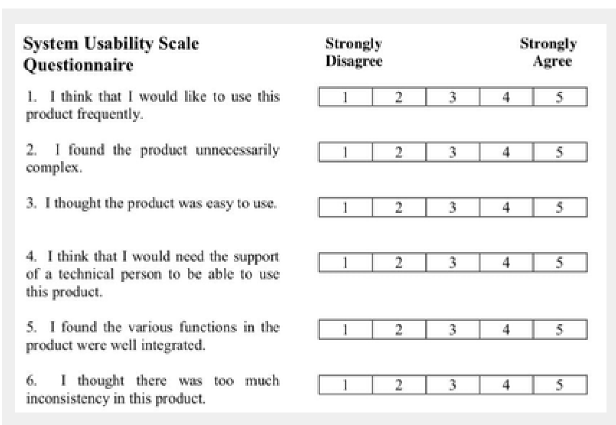
2 Objectives



- to integrate **ontology-based suggestions** on free-text annotations to support more consistent data entry;



- to develop an **Ontology Lookup Service (OLS)** to help users explore relevant ontology classes and an **Ontology Explorer Service (OES)** for examining their relationships; and



- to evaluate the system using the **System Usability Scale (SUS)**.

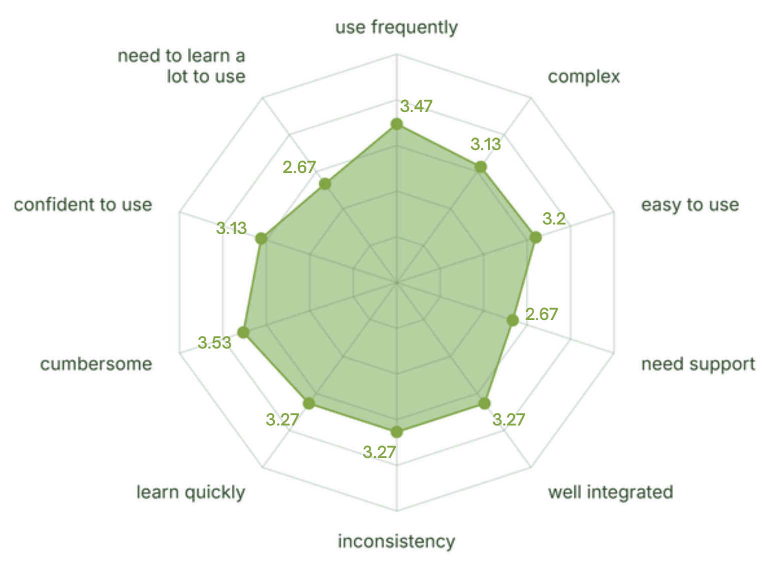
3 Methodology

The Caventology web application was developed with:

- **React.js** + **Material UI (MUI)** for the user interface,
- **Django** for the server, and **MySQL** for database management.
- The OES uses **N3.js** for parsing RDF ontology files and **Cytoscape.js** for visualization.
- The OLS and ontology-based suggestions are powered by the **BioPortal Search API**.

4 Results

- **15** respondents evaluated the system using **SUS**.
- Researchers appreciate the **tight integration** of the application's features.



- The system received a mean SUS score of **79**, indicating **good** overall usability.

5 Conclusion

By integrating ontology-based suggestions along with OLS and OES tools in the cave culture collection information system, **Caventology enhances the consistency and semantic quality of cave microbial documentation.**

Caventology

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